4.1 (d)

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$$L(S,c,\alpha) = S + \sum_{i=1}^{N} \alpha_i (\|c - x_i\|^2 - S)$$

$$= S\cdot\Big(1 - \underbrace{\sum_{i=1}^{N} \alpha_i}_{=1}\Big) + \sum_{i=1}^{N} \alpha_i \|c - x_i\|^2$$

$$= 0$$

$$= \sum_{i=1}^{N} \alpha_i \left(c^T c - 2 x_i^T c + \|x_i\|^2\right).$$

$$= \sum_{i=1}^{N} \alpha_i \|x_i\|^2 + \sum_{i=1}^{N} \alpha_i\, c^T c - \sum_{i=1}^{N} \alpha_i 2 x_i^T c$$

$$= \sum_{i=1}^{N} \alpha_i \|x_i\|^2 + \sum_{i=1}^{N} \alpha_i\, c^T c - 2 \sum_{i=1}^{N} \sum_{j=1}^{N} \alpha_i \alpha_j x_i^T x_j$$

$$= \sum_{i=1}^{N} \alpha \|x_i\|^2 + \sum_{i=1}^{N} \alpha_i \Big(\sum_{j=1}^{N} \alpha_j x_j\Big)^T \sum_{j=1}^{N} \alpha_j x_j - 2 \sum_{i=1}^{N} \sum_{j=1}^{N} \alpha_i \alpha_j x_i^T x_j$$

$$= \sum_{i=1}^{N} \alpha \|x_i\|^2 + \sum_{i=1}^{N} \sum_{j=1}^{N} \underbrace{\alpha_i^T}_{=\alpha_i} x_i^T \alpha_j x_j - 2 \sum_{i=1}^{N} \sum_{j=1}^{N} \alpha_i \alpha_j x_i^T x_j$$

$$= \sum_{i=1}^{N} \alpha \|x_i\|^2 + \sum_{i=1}^{N} \sum_{j=1}^{N} \alpha_i \alpha_j\, x_i^T x_j - 2 \sum_{i=1}^{N} \sum_{j=1}^{N} \alpha_i \alpha_j\, x_i^T x_j$$

$$= \sum_{i=1}^{N} \alpha \|x_i\|^2 - \sum_{i=1}^{N} \sum_{j=1}^{N} \alpha_i \alpha_j\, x_i^T x_i$$

$$\Rightarrow \max_{\alpha} \inf_{S,c} L(S,c,\alpha) = \max_{\alpha} \inf_{S,c} \sum_{i=1}^{N} \alpha_i \|x_i\|^2 - \sum_{i=1}^{N} \sum_{j=1}^{N} \alpha_i \alpha_j\, x_i^T$$

$$= \max_{\alpha} \sum_{i=1}^{N} \alpha_i \|x_i\|^2 - \sum_{i=1}^{N} \sum_{j=1}^{N} \alpha_i \alpha_j\, x_i^T x_i \qquad \square$$

4.1 (a)

for any c, choose S so that

$\max(\|c - x_i\|^2) = S$.

This will ensure that any point x_i is inside or on the surface of the hypersphere.

(b) $g_i(S, c) \leq 0$ with $g_i(S, c) = \|x_i + c\|^2 - S$

$\boxed{\alpha_i \geq 0}$

$$L(S, c, \alpha) = S + \sum_{i=1}^{N} \alpha_i (\|x_i + c\|^2 - S)$$
$$= S + \sum_{i=1}^{N} \alpha_i \|x_i + c\|^2 - \sum_{i=1}^{N} \alpha_i S$$

(c)

1. Primal feasibility: $\|x_i + c\|^2 - S = g_i(S, c) \leq 0$
2. Dual feasibility: $\alpha_i \geq 0$
3. Stationarity: $\partial L / \partial S = 0$, $\partial L / \partial c = 0$
4. Complementary slackness: $\alpha_i g_i(S, c) = 0$

3.: $\frac{\partial L}{\partial S} = 0 \Leftrightarrow 1 - \sum_{i=1}^{N} \alpha_i = 0 \Rightarrow \sum_{i=1}^{N} \alpha_i = 1$

$\frac{\partial L}{\partial c} = 0 \Leftrightarrow \sum_{i=1}^{N} \alpha_i 2 (c - x_i) =$

$-2c \sum_{i=1}^{N} \alpha_i + 2 \sum_{i=1}^{N} \alpha_i x_i = 0$

$-2c + 2 \sum_{i=1}^{N} \alpha_i x_i = 0 \Rightarrow c = \sum_{i=1}^{N} \alpha_i x_i$

1.: $\|x_i + c\|^2 = \|\sum_{j=1}^{N} \alpha_j x_j - x_i\|^2 \leq S$

2.: $\alpha_i \geq 0$

3.: $\alpha_i (\|c - x_i\| - S) = 0$

$\Rightarrow \alpha_i = 0$ oder $S = \|c - x_i\|$

4.1 (e) $C = \sum_{i=1}^{N} \alpha_i x_i$

$$\alpha_i \xi_i(s, C) = \alpha_i (\|C - x_i\|^2 - s)$$

$$= -\alpha_i s + \alpha_i \|C - x_i\|^2 = 0 \quad | \alpha_i \geq 0$$

$$\Leftrightarrow s = \|C - x_i\|^2 = \|\sum_{j=1}^{N} \alpha_j x_j - x_i\|^2$$

if $\alpha_i \geq 0 \rightarrow x_i$ has to stay on the hypersperes surface.